Student Information

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Answer 1

a)

We need to use the Normal approximation to determine the sample size n for the Monte Carlo simulation such that with 0.99 probability, the estimate differs from the true value by no more than 0.02.

According to our book chapter 5.3, size of a Monte Carlo study is given by:

$$N \geq p(1-p)\left(\frac{z_{\alpha/2}}{\varepsilon}\right)^2$$

Thus, the margin of error ε for a Monte Carlo simulation is:

$$\varepsilon = z_{\alpha/2}\sqrt{\frac{p(1-p)}{N}}$$

Where:

- $z_{\alpha/2}$ is the critical value from the standard normal distribution for a 99% confidence level. For 99% confidence, $\alpha = 0.01$, so $\alpha/2 = 0.005$, and $z_{\alpha/2} \approx 2.576$.
- p is the estimated probability (unknown, but we can use $p = 0.5$ for the most conservative estimate).
- N is the sample size.

We want the margin of error ε to be less than 0.02:

$$0.02 = 2.576\sqrt{\frac{0.5(1-0.5)}{N}}$$

Solving for N:

$$0.02 = 2.576\sqrt{\frac{0.25}{N}}$$

$$0.02 = 2.576 \cdot \frac{0.5}{\sqrt{N}}$$

$$0.02 = 1.288/\sqrt{N}$$

$$\sqrt{N} = 1.288/0.02$$

$$\sqrt{N} = 64.4$$

$$N \approx 64.4^2$$

$$N \approx 4147.36$$

Round up:

$$N = 4148$$

Thus, the size of the Monte Carlo simulation should be at least 4148.

b)

1. **Expected value for the weight of an automobile:**

   The weight of each automobile is Gamma distributed with parameters $\alpha = 120$ and $\lambda = 0.1$.

   The expected value $E[X]$ for a Gamma distribution is given by:

   $$E[X] = \frac{\alpha}{\lambda}$$

   For the automobile:

   $$E[\text{Weight of an automobile}] = \frac{120}{0.1} = 1200\,\text{kg}$$

2. **Expected value for the weight of a truck:**

   The weight of each truck is Gamma distributed with parameters $\alpha = 14$ and $\lambda = 0.001$.

   For the truck:

   $$E[\text{Weight of a truck}] = \frac{14}{0.001} = 14000\,\text{kg}$$

3. **Expected value for the total weights of all automobiles:**

   The number of automobiles passing over the bridge per day is a Poisson random variable with $\lambda_A = 60$.

   Since the weight of each automobile is independent and identically distributed, the total weight of all automobiles is:

   $$E[\text{Total weight of all automobiles}] = E[N_A] \cdot E[\text{Weight of an automobile}]$$

Where N_A is the number of automobiles:

$$E[\text{Total weight of all automobiles}] = 60 \cdot 1200 = 72000\,\text{kg}$$

4. **Expected value for the total weights of all trucks:**

   The number of trucks passing over the bridge per day is a Poisson random variable with $\lambda_T = 12$.

   Since the weight of each truck is independent and identically distributed, the total weight of all trucks is:

$$E[\text{Total weight of all trucks}] = E[N_T] \cdot E[\text{Weight of a truck}]$$

   Where N_T is the number of trucks:

$$E[\text{Total weight of all trucks}] = 12 \cdot 14000 = 168000\,\text{kg}$$

Answer 2

```
N = 4148;
lambdaA = 60;  % number of automobiles
lambdaT = 12;  % number of trucks

TotalWeight = zeros(1, N); % vector for Monte Carlo
for k = 1:N;
  % generate # of vehicles for Poisson
  numA = 0;
  numT = 0;


  % number of automobiles
  F = exp(- lambdaA);
  U = rand;
  while(U>=F)
    numA = numA +1;
    F = F + exp(-lambdaA) * lambdaA^numA/gamma(numA+1);
  end;

  % number of trucks
  F = exp(- lambdaT);
  U = rand;
  while(U>=F)
    numT = numT +1;
    F = F + exp(-lambdaT)*lambdaT^numT/gamma(numT+1);
  end;

  weight = 0; % total weight of vehicles for this run

  % calculate the total weight of automobiles
  % weightA = sum(gamrnd(120, 10, 1, numA));

  weightA = 0;
  for i = 1:numA
    weightA = weightA + (-(log(rand(120,1)) * 10));
  end

  % calculate the total weight of trucks
  % weightT = sum(gamrnd(14, 1000, 1, numT));

  weightT = 0;
  for i = 1:numT
    weightT = weightT + (-(log(rand(14,1)) * 1000));
  end

  TotalWeight(k) = sum(weightA) + sum(weightT);

end;

p_est = mean(TotalWeight > 250000);
expectedWeight = mean(TotalWeight);
stdWeight = std(TotalWeight);

fprintf('Estimated probability = %f\n',p_est);
fprintf('Expected weight = %f\n',expectedWeight);
fprintf('Standard deviation = %f\n',stdWeight);
```

Figure 1: The Code for Simulation

```
Estimated probability = 0.405256
Expected weight = 240196.669097
Standard deviation = 51459.483624
```

Figure 2: Expected Output of the Code

From our simulation, according to Figure 2, we have found that the probability of the bridge collapsing is 0.405, indicating a significant risk. Additionally, the total weight of all vehicles crossing the bridge was estimated to be approximately 240196. The standard deviation of this total weight was estimated to be approximately 51459 which is relatively high. This indicates significant variability in the daily total weights observed in the simulation. A high standard deviation implies that the weights of the vehicles passing over the bridge can fluctuate widely from day to day. Consequently, while our estimate of the expected total weight is based on a large sample size and is statistically sound, the substantial variability reflected by the high standard deviation reduces the precision of this estimate.

To find the new Poisson variable λ_T so that the probability of bridge's collapse is less than 0.1, we can change manually the variable (decrease one by one) and rerun our simulation.

Let's decrease our λ_T by 1 ($\lambda_T = 11$):

```
Estimated probability = 0.286403
Expected weight = 224589.189491
Standard deviation = 47463.984548
```

Figure 3: Simulation result with $\lambda_T = 11$

We get approximately 0.29, so keep going until go below 0.1 :

```
Estimated probability = 0.200579
Expected weight = 211981.603840
Standard deviation = 46826.382459
```

Figure 4: Simulation result with $\lambda_T = 10$

```
Estimated probability = 0.129942
Expected weight = 198226.802187
Standard deviation = 44853.292178
```

Figure 5: Simulation result with $\lambda_T = 9$

```
Estimated probability = 0.067502
Expected weight = 183503.096615
Standard deviation = 42266.100893
```

Figure 6: Simulation result with $\lambda_T = 8$

Now we get the probability as 0.06 which is less then 0.1. Thus, we can say that the highest limit for number of trucks is 8.